MISSION: QUICK AID

murusSG

Multi-source Unified Response and Urgency System

THE FRAGMENTATION CRISIS

In crisis response, **silos cause seconds of lag** that become minutes of delay.

· **Disjointed Agency Telemetry:** Responders juggle between SGSecure, myResponder, and various agency feeds.

· **Isolated Command Centers:** Information stays locked within individual agency dashboards.

· **High-Noise Intelligence:** Citizens receive raw text instructions without actionable spatial context.

Singapore skyline with network overlay

Fragmented Data Costs Lives

Eliminating the chaos of disjointed systems to empower synchronised, lifesaving decisions

ONE SINGLE SOURCE OF TRUTH

Unified Telemetry

Fusing real-time agency data from several agencies like NEA, PUB, and LTA into one dynamic layer.

Crowd Intelligence

Integrating crowd-sourced field reports with resource capacity mapping.

Dynamic Mapping

A unified ecosystem eliminating the need to switch between disjointed systems.

INTEGRATED INGESTION LAYER

Agency Source	Feeds Ingested	Operational Impact
NEA	PSI, PM2.5, Lightning, Dengue	Environmental Hazard Awareness
PUB	Water Level Sensors, Flood Alerts	Flash Flood Prevention
MOH	DORSCON, Infectious Disease Bulletins	Pandemic & Bed Load Tracking
SCDF & SPF	myResponder Incidents, Road Safety	Rapid Community Response
LTA	Traffic Cams, MRT Disruptions	Evacuation Route Optimization

Normalization: Every data point is converted into a unified event schema for role-specific decision support.

GEOSPATIAL EVENT ENGINE

Contextual Intelligence

Every incoming data point is automatically tagged with 4 key parameters to power the **"Is this for me?"** filter:

· **Location:** Precise GPS coordinates via OneMap API. Location will be tagged to the location user is in if location service is turned on, else it defaults to home address set by user

· **Severity:** Real-time hazard level classification.

· **Hazard Type:** Biological, Environmental, or Security.

· **Vicinity Radius:** Street-level (Dengue), Regional (Haze), Hyperlocal (Flood).

VISUALIZED ACTIONABLE ALERTS

Actionable Steps

Converting raw text instructions into visual guides that the general community can follow instantly.

Evacuation Routing

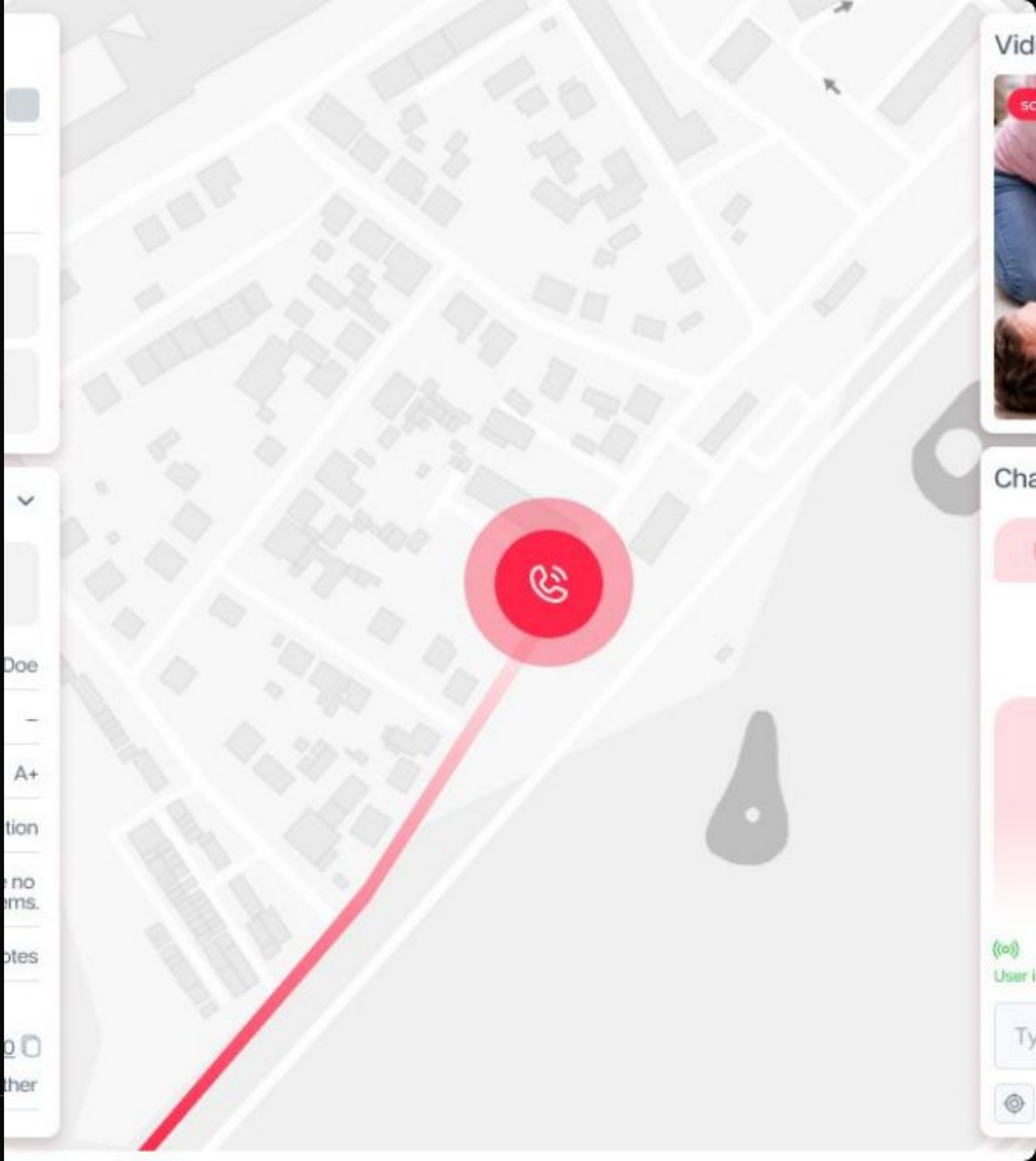
Visualizing proximity hazards to provide clear, safe evacuation routes during distress situations.

OneMap Integration

Leveraging the OneMap API to broadcast geospatial guidance directly to resident devices.

ACTIONABLE
GUIDANCE

By integrating the **OneMap API**, murusSG transforms plain text advisories into a geospatial experience.
Residents no longer guess if a flood warning applies to their block—they see their proximity, the safest path, and active responder locations in real-time.



Shared Incident Management

Eliminating transparency gaps and resource overlaps between SPF, SCDF, and other government agencies involved

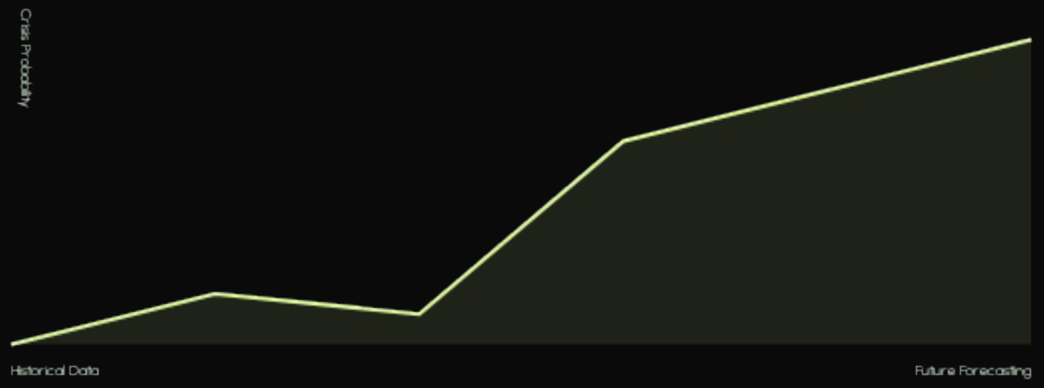
ELIMINATING SILOS

· **Transparent Relay:** Information flows top-to-bottom for ALL public safety incidents.

· **Minimal Data Lag:** Automated dispatching of resources based on the closest available agency assets with human approval in place..

· **Unified Dashboard:** A shared system for responders to coordinate operations without disjointed updates.

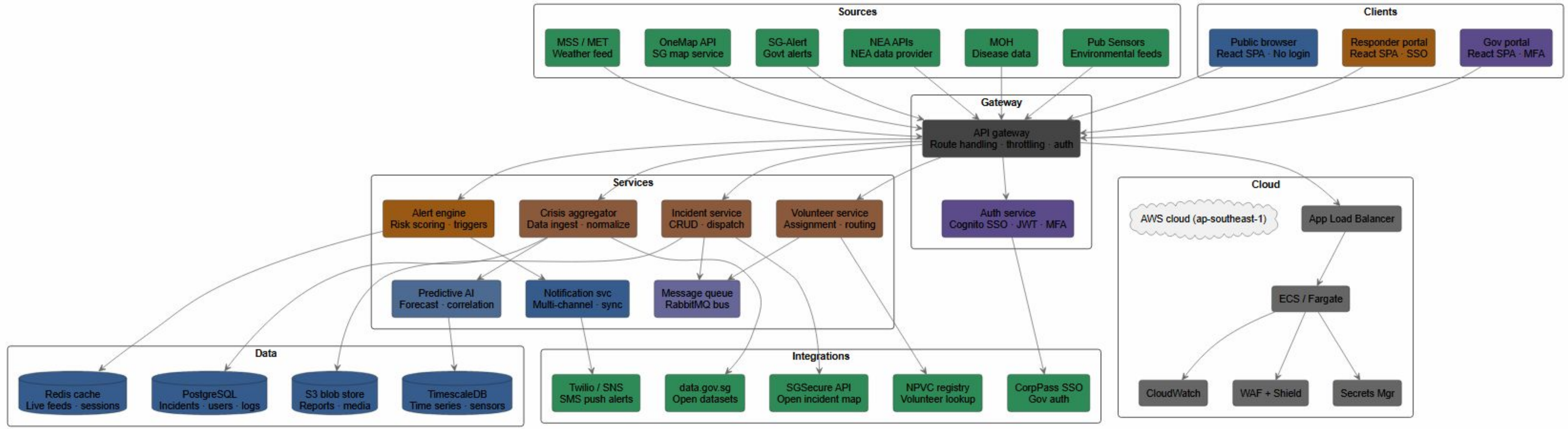
PREDICTIVE INTELLIGENCE



Forecasting Hotspots
Leveraging historical trends to predict crisis escalation before it happens.

Enabling leaders to pre-position resources and medical assets in potential high-demand zones, transforming response from **Reactive to Proactive**.

Architecture powering murusSG



Color	Meaning
Green	External data source / integration
Blue	Frontend or database
Orange	Backend domain service
Purple	Auth / identity